

Linear Regression

Complete Study Notes

1. What is Linear Regression?

Linear Regression is a supervised machine learning algorithm used for predictive analysis. It determines the linear relationship between the dependent variable (Y) and independent variable(s) (X). This relationship is represented by a straight line called the **regression line / best-fit line**.

It is used to predict a **quantitative/continuous output**, such as age, salary, price, sales, etc.

2. Simple Linear Regression

Simple Linear Regression has **one X variable and one Y variable**.

Formula:

$$y = mx + c$$

y = response or target variable

x = predictor variable

m = coefficient/slope of x

c = intercept

3. Multiple Linear Regression

Multiple Linear Regression has **one Y variable and two or more X variables**.

Formula:

$$y = m_1x_1 + m_2x_2 + \dots + m_nx_n + c$$

y = target variable

x₁, x₂, ..., x_n = features

m₁, m₂, ..., m_n = coefficients of the respective features

c = intercept

4. Cost Function

A **cost function** signifies how much the predicted values are deviated from the actual values.

MSE (Mean Squared Error) is one of the most commonly used error/cost measures for Linear Regression. It is the average of the squared differences between predicted and actual values. Its output is a single number representing the cost.

$$MSE = (1/N) \sum (y_i - \hat{y}_i)^2$$

5. Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the cost function. It iteratively adjusts model parameters to move toward the minimum of the cost function.

Our aim is to **minimize the error between predicted values and actual values**.

The algorithm starts with an initial value for the parameter (such as the slope m), calculates the cost, and repeatedly adjusts the parameter to search for a minimum error. Once the minimum is reached, the resulting parameters give the best-fit line.

The **learning rate** controls the size of each step. A very large learning rate can overshoot the minimum, while a very small learning rate can make training unnecessarily slow.

Basic update idea:

$$\text{new parameter} = \text{old parameter} - \text{learning rate} \times \text{gradient}$$

6. Model Evaluation

Model evaluation is important in Data Science because it helps us understand model performance and compare models.

7. R^2 (R-Squared) Statistics

R^2 measures how much of the variability in the dependent variable can be explained by the model. It is a measure of goodness of fit.

For example, if $R^2 = 0.75$, the model explains approximately **75% of the variation** in the target variable.

Formula:

$$R^2 = (TSS - RSS) / TSS = 1 - RSS/TSS$$

8. RSS and TSS

RSS (Residual Sum of Squares) represents the remaining prediction error:

$$RSS = \sum (y_i - \hat{y}_i)^2$$

TSS (Total Sum of Squares) represents the total variation of Y around its mean:

$$TSS = \sum (y_i - \bar{y})^2$$

Therefore, R^2 compares the model's remaining error (RSS) with the total variation (TSS). A higher R^2 generally indicates a better fit.

Important: R^2 does not mean that a certain percentage of individual data rows are correct. It represents the proportion of variation in the target explained by the model.

9. Adjusted R^2

Ordinary R^2 generally increases or stays the same when additional predictors are added, even when those predictors are not useful. This can make R^2 misleading in Multiple Linear Regression.

Adjusted R^2 addresses this by penalizing the addition of unnecessary predictors.

$$\text{Adjusted } R^2 = 1 - [(1 - R^2)(N - 1) / (N - p - 1)]$$

R^2 = sample R-squared

p = number of predictors

N = total sample size

Adjusted R^2 is always less than or equal to R^2 for the standard regression setup, and it can decrease when an added predictor does not provide enough useful information.

10. MSE and RMSE

MSE is an absolute measure of prediction error based on squared errors:

$$\text{MSE} = (1/N) \sum (y_{\text{actual}} - y_{\text{predicted}})^2$$

It gives a numerical measure of how much predictions deviate from actual values. Because errors are squared, large errors are penalized more strongly.

RMSE (Root Mean Squared Error) is the square root of MSE:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

RMSE is commonly easier to interpret because it is expressed in the same units as the target variable.

11. Mean Absolute Error (MAE)

MAE is similar to MSE, but instead of squaring the errors it uses their absolute values.

$$\text{MAE} = (1/N) \sum |y_{\text{actual}} - y_{\text{predicted}}|$$

MAE represents the average absolute difference between actual and predicted values.

12. Quick Comparison

Metric	Meaning	Better Value
MAE	Average absolute prediction error	Lower
MSE	Average squared prediction error	Lower
RMSE	Square root of MSE; same units as target	Lower
R^2	Proportion of target variation explained by model	Higher
Adjusted R^2	R^2 adjusted for number of predictors	Higher

13. Complete Mental Model

Linear Regression → learns the relationship between X and Y

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Equation → $y = mx + c$ (simple) or $y = m_1x_1 + \dots + m_nx_n + c$ (multiple)

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Prediction → model predicts Y

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Error / Cost → compare predicted values with actual values

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Gradient Descent → minimizes the cost by adjusting parameters

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Model Evaluation → R^2 , Adjusted R^2 , MSE, RMSE, MAE

14. Interview / Exam Quick Answers

What is Linear Regression?

A supervised machine learning algorithm used to predict a continuous numerical target by modeling a linear relationship between predictors and the target.

What is Gradient Descent?

An optimization algorithm that iteratively updates model parameters to minimize the cost function.

What is R^2 ?

A metric that measures the proportion of variation in the dependent variable explained by the regression model.

Why Adjusted R^2 ?

It accounts for the number of predictors and penalizes unnecessary variables.

Which is better: lower or higher error?

For MAE, MSE and RMSE, lower is better. For R^2 and Adjusted R^2 , higher is generally better.